WORKED SOLUTION

STEP II 2019, Question 8

Pure • Multiplicative Matrix Functions

STEP II 2019, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 7.3, *What Makes a Determinant*: https://beyondstep.keithmcnulty.org/ch07/determinant-axioms.html

The key idea

Everything follows from the single rule $\mathrm{f}(\mathbf{MN}) = \mathrm{f}(\mathbf{M})\mathrm{f}(\mathbf{N})$, applied to well-chosen products. The trick each time is to find two different ways of writing the same matrix, or a matrix that is a product of itself with something else, and then compare the values of f. The determinant is one such function, so it's a useful guide to what the answers ought to be, but we must not assume that f *is* the determinant.

Solution

Part (i)

Since $\mathbf{MI} = \mathbf{M}$,

$$\mathrm{f}(\mathbf{M}) = \mathrm{f}(\mathbf{MI}) = \mathrm{f}(\mathbf{M})\,\mathrm{f}(\mathbf{I}).$$

We are told $\mathrm{f}(\mathbf{M}) \neq 0$, so we may divide by it: $\mathrm{f}(\mathbf{I}) = 1$.

(Without that assumption we could not conclude this: the function that is identically zero is multiplicative too.)

Part (ii)

Multiplying out, $\mathbf{J}^2 = \mathbf{I}$, so

$$\mathrm{f}(\mathbf{J})^2 = \mathrm{f}(\mathbf{J}^2) = \mathrm{f}(\mathbf{I}) = 1.$$

Hence $\mathrm{f}(\mathbf{J}) = \pm 1$, and since $\mathrm{f}(\mathbf{J}) \neq 1$ we get $\mathrm{f}(\mathbf{J}) = -1$.

? **Ask yourself** What does multiplying by $\mathbf{J}$ do to a matrix?

Multiplying on the left by $\mathbf{J}$ swaps the rows; multiplying on the right swaps the columns:

$$\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}a&b\\c&d\end{pmatrix}=\begin{pmatrix}c&d\\a&b\end{pmatrix},\qquad \begin{pmatrix}c&d\\a&b\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}=\begin{pmatrix}d&c\\b&a\end{pmatrix}.$$

Applying f to each and using $\mathrm{f}(\mathbf{J})=-1$,

$$\mathrm{f}\begin{pmatrix}c&d\\a&b\end{pmatrix}=-\mathrm{f}\begin{pmatrix}a&b\\c&d\end{pmatrix},\qquad \mathrm{f}\begin{pmatrix}d&c\\b&a\end{pmatrix}=-\mathrm{f}\begin{pmatrix}c&d\\a&b\end{pmatrix}.$$

The first gives $\mathrm{f}\begin{pmatrix}a&b\\c&d\end{pmatrix}=-\mathrm{f}\begin{pmatrix}c&d\\a&b\end{pmatrix}$, and the second, multiplied by -1, then shows this also equals $\mathrm{f}\begin{pmatrix}d&c\\b&a\end{pmatrix}$, as required.

Part (iii)

Let the first row of $\mathbf{A}$ be $(a\ \ b)$ and the second row λ times it. Let $\mathbf{B}$ be the matrix with two equal rows, and take $k=\lambda$ in $\mathbf{K}$. Multiplying on the left by $\mathbf{K}$ scales the second row by k, so

$$\mathbf{A}=\begin{pmatrix}a&b\\\lambda a&\lambda b\end{pmatrix}=\begin{pmatrix}1&0\\0&\lambda\end{pmatrix}\begin{pmatrix}a&b\\a&b\end{pmatrix}=\mathbf{KB}.$$

Now swapping the rows of $\mathbf{B}$ leaves it unchanged, so by part (ii) (with $c=a$, $d=b$)

$$\mathrm{f}(\mathbf{B})=-\mathrm{f}(\mathbf{B}),$$

which forces $\mathrm{f}(\mathbf{B})=0$. Therefore $\mathrm{f}(\mathbf{A})=\mathrm{f}(\mathbf{K})\,\mathrm{f}(\mathbf{B})=0$. This works for every real λ, including $\lambda=0$.

Part (iv)

First compute the three matrices. For $k\neq 0$, $\mathbf{K}^{-1}=\begin{pmatrix}1&0\\0&1/k\end{pmatrix}$, and

$$\mathbf{P}^2=\begin{pmatrix}1&2\\0&1\end{pmatrix},\qquad \mathbf{P}^{-1}=\begin{pmatrix}1&-1\\0&1\end{pmatrix},\qquad \mathbf{K}^{-1}\mathbf{P}\mathbf{K}=\begin{pmatrix}1&0\\0&1/k\end{pmatrix}\begin{pmatrix}1&k\\0&k\end{pmatrix}=\begin{pmatrix}1&k\\0&1\end{pmatrix}.$$

Next, conjugating by $\mathbf{K}$ does not change f. Indeed $\mathrm{f}(\mathbf{K}^{-1})\mathrm{f}(\mathbf{K})=\mathrm{f}(\mathbf{K}^{-1}\mathbf{K})=\mathrm{f}(\mathbf{I})=1$, so

$$\mathrm{f}(\mathbf{K}^{-1}\mathbf{P}\mathbf{K})=\mathrm{f}(\mathbf{K}^{-1})\,\mathrm{f}(\mathbf{P})\,\mathrm{f}(\mathbf{K})=\mathrm{f}(\mathbf{P}).$$

? **Ask yourself** Which values of k turn $\mathbf{K}^{-1}\mathbf{P}\mathbf{K}$ into $\mathbf{P}^2$ and $\mathbf{P}^{-1}$?

Taking $k=2$ gives $\mathbf{K}^{-1}\mathbf{P}\mathbf{K}=\mathbf{P}^2$, so

$$\mathrm{f}(\mathbf{P})=\mathrm{f}(\mathbf{P}^2)=\mathrm{f}(\mathbf{P})^2,$$

and so $\mathrm{f}(\mathbf{P})=0$ or 1.

Taking $k = -1$ gives $\mathbf{K}^{-1}\mathbf{P}\mathbf{K} = \mathbf{P}^{-1}$, so $\mathrm{f}(\mathbf{P}^{-1}) = \mathrm{f}(\mathbf{P})$. But $\mathrm{f}(\mathbf{P})\,\mathrm{f}(\mathbf{P}^{-1}) = \mathrm{f}(\mathbf{I}) = 1$, so

$$\mathrm{f}(\mathbf{P})^2 = 1,$$

and in particular $\mathrm{f}(\mathbf{P}) \neq 0$. Combining the two results,

$$\mathrm{f}(\mathbf{P}) = 1.$$

Checks

The determinant satisfies every hypothesis: it is multiplicative, $\det \mathbf{I} = 1$ and $\det \mathbf{J} = -1$. It agrees with all our conclusions: swapping rows or columns changes its sign, a matrix with proportional rows has determinant 0, and $\det \mathbf{P} = 1$. A less obvious example is $\mathrm{f}(\mathbf{M}) = \operatorname{sgn}(\det \mathbf{M})\,|\det \mathbf{M}|^{0.7}$. A computer check on thousands of random matrices confirmed the multiplicative rule, the row and column swap identities and $\mathrm{f}(\mathbf{A}) = 0$ for proportional rows, as well as $\mathrm{f}(\mathbf{P}) = 1$, for both functions.

Common slips

- Assuming $\mathrm{f} = \det$. The question asks you to deduce the properties from multiplicativity alone; the determinant is only a guide.
- Using $\mathrm{f}(\mathbf{K}^{-1}) = 1/\mathrm{f}(\mathbf{K})$ without justification. It needs $\mathrm{f}(\mathbf{I}) = 1$ from part (i) and $k \neq 0$, so that $\mathbf{K}$ is invertible.
- Stopping at $\mathrm{f}(\mathbf{P})^2 = \mathrm{f}(\mathbf{P})$ and forgetting to rule out $\mathrm{f}(\mathbf{P}) = 0$.

Beyond STEP

A map such as f that respects multiplication is a *homomorphism* from the group of invertible 2×2 matrices to the non-zero reals. Part (iv) is the start of a classic argument: the shear matrices like $\mathbf{P}$ are products of commutators, so every such homomorphism sends them to 1, and f ends up depending only on the determinant. You can also show that the determinant is the only function that is linear in each row, alternating, and equal to 1 at $\mathbf{I}$. Strang's *Introduction to Linear Algebra* builds the determinant from properties like these, and Allenby's *Rings, Fields and Groups* covers homomorphisms.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.